

Medical Oncology, New York, NY, <sup>9</sup>Department of Medical Oncology, Memorial Sloan Kettering Cancer Center, New York, NY

**Purpose/Objective(s):** To analyze clinical and dosimetric factors for the occurrence of temporal lobe necrosis (TLN) among patients who received proton re-irradiation therapy involving the skull base.

**Materials/Methods:** We analyzed 99 consecutive patients who received proton re-irradiation involving the skull base between 09/2013 and 03/2017. Follow up interval was defined from the beginning of proton therapy. Analyzed factors included gender, history of hypertension and/or diabetes, smoking status, prior radiation dose, composite dose to 2cc and composite max pixel dose to the temporal lobe. Outcomes of interest included the grade of TLN, incidence as well as time to development of TLN among patients treated in the re-irradiation setting. Median doses were compared using the Mann-Whitney's test. Univariate and multivariate Cox regression analysis were performed.

**Results:** Of the 99 patients, 86 (87%) were treated for recurrent locoregional disease and 12 (12%) for metastatic disease. The median follow up was 12 months (IQR 6-21) for the entire cohort and 17 months (IQR 9-26) for living patients; median survival at 1 year was 64.5%. The most common primary sites were the nasopharynx (n=22) and paranasal cavity/sinuses (n=18) and the most common histologies included squamous cell carcinoma (n=51) and adenoid cystic carcinoma (n=12). The median first course radiation dose to the skull base was 6360cGy for those without TLN and 7020cGy for those with TLN (p=0.02). Median re-irradiation proton dose was 60CGE without TLN and 70CGE with TLN (p=0.18). The average interval between first and second course treatment was 25 months. 15 (15%) developed temporal lobe necrosis within a median time of 19 months (IQR 15-30); 9 had right-sided, 5 had left-sided, and 1 had bilateral TLN. The median composite max point dose was 63.7Gy among patients without TLN and 120Gy for those with TLN (p<0.001). The median composite dose to 2cc was 73.1Gy among patients without TLN and 134.5Gy with TLN (p<0.001). On multivariate Cox regression, male gender (HR 8.47, p=0.008), hypertension (HR 4.78, p=0.017) and higher composite max point dose (HR 1.03, p=0.011) were significantly associated with development of TLN. All patients were determined to have grade 1 TLN per CTCAE 4.0 that was found on imaging and were clinically asymptomatic except for two patients. Both required medical management with steroids.

**Conclusion:** Among patients treated in the re-irradiation setting with proton therapy, 15% of patients developed TLN, which is comparable to TLN rates in the upfront setting with IMRT. The rate of grade 3 TLN was 1% and there was no grade 4 TLN. No TLN was observed in patients with a composite temporal lobe D2cc under 60Gy, or with a composite max point dose under 72Gy.

**Author Disclosure:** A. Lee: None. K.L. Pitter: None. B. Neal: None. C. Chow: None. K. Sine: None. O. Cahlon: None. S. McBride: None. J.E. Leeman: None. N. Riaz: None. T. Waldenberg: None. M. Cohen: None. I. Ganly: None. J.O. Boyle: None. R.J. Wong: None. C.W. Brennan: None. E. Sherman: None. L. Michel: None. N. Lee: Consultant; Lily. Advisory Board; Pfizer, Vertex, Merck. C. Tsai: None.

## M0\_25\_2696

### A Contour-Based Approach for Predicting Corneal Toxicity in Patients with Uveal Melanoma Treated with Proton Therapy



H.J. Lee, Jr.<sup>1</sup> A. Stacey,<sup>2</sup> T. Klesert,<sup>3</sup> C. Wells,<sup>3</sup> A. Skalet,<sup>4</sup> C. Bloch,<sup>5</sup> A. Fung,<sup>6</sup> S.R. Bowen,<sup>5,7</sup> T.P. Wong,<sup>6</sup> D. Shibata,<sup>7</sup> L.M. Halasz,<sup>5</sup> and R. Rengan<sup>5</sup>; <sup>1</sup>Duke University School of Medicine, Durham, NC, <sup>2</sup>University of Washington, Department of Ophthalmology, Seattle, WA, <sup>3</sup>Vitreoretinal Associates of Washington, Seattle, WA, <sup>4</sup>Oregon Health and Science University, Portland, OR, <sup>5</sup>University of Washington, Department of Radiation Oncology, Seattle, WA, <sup>6</sup>SCCA Proton Therapy Center, Seattle, WA, <sup>7</sup>University of Washington, Department of Radiology, Seattle, WA

Abstract M0\_25\_2696; Table 1

| Limbus Parameter  | Toxicity median (n=8) | No Toxicity median (n=63) | Cutoff | ROC AUC | P-value |
|-------------------|-----------------------|---------------------------|--------|---------|---------|
| V5 %              | 100                   | 58.29                     | 79.27  | 0.80    | 0.007   |
| V20 %             | 100                   | 4.17                      | 36.96  | 0.91    | <0.001  |
| V45 %             | 48.16                 | 0                         | 3.95   | 0.96    | <0.001  |
| V50 %             | 26.74                 | 0                         | 0.41   | 0.91    | <0.001  |
| Mean Dose Gy(RBE) | 41.90                 | 10.40                     | 17.16  | 0.95    | <0.001  |
| Max Dose Gy(RBE)  | 52.47                 | 30.00                     | 48.29  | 0.97    | <0.001  |

**Purpose/Objective(s):** Corneal toxicity is a complication of proton beam radiation therapy (PBT) for uveal melanoma, especially in anteriorly-positioned tumors. However, in the previous era of model-based treatment planning, the corneal dose was difficult to accurately quantify. Treatment planning today has shifted to contour-based planning from a model-based approach, allowing for dose-volume histogram (DVH) analysis of the cornea and development of optimization approaches to minimize the risk of this toxicity. The primary objective of this study was to examine comparative corneal substructure doses in patients who experienced corneal toxicity with those who did not in order to develop dose constraints.

**Materials/Methods:** The medical records of 71 patients with uveal melanoma who received PBT at a single institution (either 50 Gy(RBE) in 5 fractions or 56 Gy(RBE) in 4 fractions) were retrospectively reviewed with institutional review board approval and analyzed for toxicity. Minimum follow-up was 6 months. 36 patients, including all patients with anterior tumors for whom MR images were available for contouring, were included in the DVH analysis. All patients were treated using uniform scanning technique. Corneal contours for DVH analysis were generated in a commercially available deformable registration algorithm using 3D T2-weighted MRI of the orbit co-registered to a treatment planning CT scan. The cornea was defined as a 12 mm diameter contour along the anterior surface of the eye, extending to the limbus, defined as the inflection point between the corneal and scleral curvatures. ROC analysis and Mann-Whitney testing with Benjamini-Hochberg false discovery rate correction were used to predict incidence of corneal toxicity. We also report cutoff values that maximize the Youden Index = sensitivity + (specificity minus one) for predicting toxicity incidence and are most applicable as planning constraints.

**Results:** All eight patients experiencing toxicity had anteriorly positioned tumors, with tumor extension anterior to the equator of the globe. PTV volume dichotomized by the median did not correlate with corneal toxicity (Fisher p=0.16). Median PTV D95% was 50.9 Gy(RBE) for both groups (p=0.89). Mean dose, max dose, V5, V20, V45, and V50 for the limbus were all significantly higher in patients experiencing corneal toxicity (p≤0.007). Candidate DVH cutoff parameters for the corneal V5, V20, V45, V50, as well as mean and max corneal doses are given in the table. All cases of toxicity occurred within 1-6 months of follow up, at a median follow up of 3 months.

**Conclusion:** Corneal toxicity is a notable side effect of PBT for anteriorly positioned uveal melanoma, and dosimetry can aid prediction of patients at risk for toxicity who may benefit from early intervention.

**Author Disclosure:** H.J. Lee: None. A. Stacey: None. T. Klesert: None. C. Wells: None. A. Skalet: None. C. Bloch: Oversee the organization of online continuing educational material and related duties; American Association of Medical Physicist. A. Fung: None. S.R. Bowen: Research Grant; National Cancer Institute. T.P. Wong: None. D. Shibata: None. L.M. Halasz: Research Grant; Fred Hutch/Univ of Washington Cancer Consortium. R. Rengan: Honoraria; Apollo Oncology Group. Consultant; Apollo Oncology Group, Astra-Zeneca. Advisory Board; Apollo Oncology Group. Travel Expenses; Apollo Oncology Group.